

# Federated Multi-Agent AI for Privacy-Preserving, Energy-Aware Optimisation of Public Transport Systems

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## Abstract

This paper presents a Federated Multi-Agent Artificial Intelligence (FMA-AI) framework for energy-aware optimisation of urban public transport under data-sovereignty constraints. Route-level agents learn a shared speed-regulation policy through synchronous federated averaging (FedAvg), so that only model parameters, and never operational data, leave each agent. The local objective minimises traction, auxiliary and congestion-related energy subject to a bounded loss in on-time performance (OTP). The framework is evaluated in a simulation built on the Massachusetts Bay Transportation Authority (MBTA) GTFS feed (summer 2025), using the six most frequent weekday bus routes as training agents and 24 further routes as a hold-out set, with stochastic arrival deviations generated by a surrogate model and results aggregated over 20 random seeds. Under a punctuality tolerance of 2 percentage points, FedAvg reduces energy consumption and CO<sub>2</sub> emissions by 0.78% (95% CI  $\pm 0.07$ ), lowering normalised energy from 0.1000 to 0.0992 kWh per passenger-kilometre, while OTP falls from 71.2% to 70.1%. This outcome is statistically indistinguishable from a centralised optimiser that pools all data ( $0.83\% \pm 0.08$ ; paired difference  $-0.05 \pm 0.11$  points), and the shared policy transfers to the 24 unseen routes with a 0.88% reduction, which route-isolated learners cannot achieve. A tolerance sweep indicates an approximately linear trade-off of about 0.66 percentage points of energy per percentage point of OTP, and no energy saving is attainable without some loss of punctuality under the adopted surrogate. FedAvg met a moving-average convergence criterion after a median of 215 communication rounds. The findings suggest that federated learning can approach centralised performance in transit control without data pooling, while the magnitude of achievable savings from speed regulation alone appears modest.

**Keywords:** Federated Learning; Multi-Agent Reinforcement Learning; Public Transport Optimisation; Energy-Aware Control; Privacy Preservation; GTFS; Sustainable Urban Mobility.

## 1. Introduction

Urban public transport sits at the intersection of two pressures that are often treated separately: the need to decarbonise operations and the need to maintain reliable service in increasingly complex networks. Conventional planning approaches built on linear forecasting and deterministic scheduling have been described as poorly suited to the variability of contemporary urban mobility (Saki & Soori, 2025). Artificial intelligence, and in particular deep and reinforcement learning, has accordingly been proposed as a means of modelling flows, adapting schedules and simulating operational scenarios (Bharadiya, 2023), and deep reinforcement learning has been applied to online multimodal transportation planning (Farahani et al., 2021).

A parallel strand of work stresses the environmental dimension. Urban mobility is energy-intensive, and AI-based optimisation that incorporates sustainability criteria has been reviewed as a promising route towards

more sustainable transport and logistics operations (Chen et al., 2024; Nikitas et al., 2020). Large language models and generative AI have more recently been discussed as tools for participatory planning and scenario exploration (Da et al., 2025; Yan et al., 2025), although their role in real-time control remains largely conceptual.

Two obstacles limit the practical uptake of these methods. First, many optimisation schemes assume centralised access to operational data, which raises privacy and data-sovereignty concerns when several operators or depots are involved. Second, in our reading of the literature, many studies rely on synthetic networks, which weakens external validity, and energy and emissions are often computed after the fact rather than embedded in the control objective.

This paper examines whether federated learning can reconcile these requirements in a deliberately transparent setting. The contribution is threefold. First, it specifies an energy-aware control objective in which energy is minimised subject to an explicit, operator-chosen bound on punctuality loss, rather than through ad hoc weightings. Second, it compares centralised, route-isolated and federated learning regimes under identical data, compute budgets and stochastic conditions, with uncertainty quantified over 20 seeds and generalisation assessed on routes excluded from training. Third, it reports the energy and punctuality trade-off as a frontier, so that the modest magnitude of achievable savings is made explicit rather than obscured by a single operating point. The empirical setting is a simulation built on the public MBTA GTFS feed; the stochastic deviation model is synthetic, and the results should therefore be read as evidence about the relative behaviour of the learning regimes rather than as field-validated savings.

## **2. Related Work**

Multi-agent systems have been proposed for distributed, event-driven traffic management, with emphasis on scalability across large networks (Horn et al., 2020). Within public transport specifically, multi-agent reinforcement learning (MARL) has been applied to holding control for the prevention of bus bunching, where each bus acts as an agent and a shared reward encourages regular headways (Wang & Sun, 2020). Advances in MARL coordination have also been reported in other control domains, such as water environment systems (Jia & Pei, 2025), which suggests that the underlying techniques are transferable, although their adaptation to transit operations is not automatic.

Federated learning (FL) trains a shared model across clients that retain their data locally. The canonical FedAvg algorithm aggregates locally updated parameters through a sample-weighted average (McMahan et al., 2017). When client objectives differ, local updates drift towards client-specific optima, which can slow convergence and bias the aggregate; control-variate corrections such as SCAFFOLD have been proposed to mitigate this effect (Karimireddy et al., 2020). In intelligent transportation systems, FL has been applied mainly to traffic flow prediction, perception and edge processing (Zhang et al., 2024). Asynchronous FL schemes have been developed for smart public transportation to reduce aggregation latency (Xu et al., 2023), multi-objective FL has been used for traffic prediction in vehicular networks (Aalavanthar et al., 2025), and federated reinforcement learning has been shown to scale traffic signal control across intersections (Bao et al., 2023).

Generative AI has been surveyed as a support for descriptive, predictive and simulation tasks in transportation planning, with open challenges concerning data scarcity, bias and alignment with sustainability goals (Da et al., 2025). Conceptual architectures combining large language models and retrieval-augmented generation with multi-agent intelligent transportation systems have also been proposed (Xu et al., 2024).

Table 1 summarises these strands and the gaps that motivate the present study.

**Table 1. Identified gaps**

| Area                             | Core contributions                                                     | Remaining gaps                                                                    |
|----------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Multi-agent systems and MARL     | Distributed control; MARL holding control for bus bunching             | Energy is rarely part of the objective; privacy of operator data is not addressed |
| Federated learning               | Privacy-preserving training; asynchronous and multi-objective variants | Applied mostly to prediction rather than operational control                      |
| Federated reinforcement learning | Scalable federated control of traffic signals                          | Limited application to public transport operations and energy objectives          |
| Generative AI and LLMs           | Planning support and multi-agent advisory concepts                     | Limited application to real-time, decentralised transit control                   |

To our knowledge, few studies have jointly examined (i) a GTFS-based transit network, (ii) route-level learning agents, (iii) federated aggregation without data pooling, and (iv) an objective that embeds energy while constraining punctuality loss. The present study addresses this combination at a deliberately small scale, with the aim of establishing whether federation preserves the performance of centralised optimisation and how large the attainable energy savings are under transparent assumptions.

### 3. Methodology

#### 3.1 Data and route selection

The study uses the MBTA GTFS archive (feed version “Summer 2025, version D”, published 7 August 2025, valid from 31 July to 23 August 2025). Although this version was published one day after the service day analysed, it covers the full validity window from 31 July and therefore contains the planned schedule for that day. Active services are resolved for a representative weekday, Wednesday 6 August 2025, from `calendar.txt` and the exceptions in `calendar_dates.txt`. Bus routes (`route_type` = 3, which in the MBTA feed includes the Silver Line bus rapid transit routes) are ranked by the number of trips scheduled on that day. The six most frequent routes form the training agents and the next 24 routes form a hold-out set that is never used for training.

For each route, the representative trip is selected among direction-0 trips that follow the most common shape on the service day, taking the trip whose first departure is closest to 12:00. Its route length  $L_i$  is the great-circle (haversine) length of the shape polyline in `shapes.txt`, its scheduled running time  $T_i$  is the difference between the last arrival and the first departure, and its stop schedule  $t_{ij}$  is taken from `stop_times.txt`. The number of daily trips  $n_i$  is the count of all trips of the route on the service day. Table 2 lists the training routes.

**Table 2. Training routes (MBTA GTFS, Wednesday 6 August 2025)**

| Route | Trips per day $n_i$ | Length $L_i$ (km) | Running time $T_i$ (min) | Stops |
|-------|---------------------|-------------------|--------------------------|-------|
| 111   | 377                 | 9.00              | 27                       | 25    |
| 32    | 289                 | 7.32              | 22                       | 25    |
| 116   | 246                 | 8.80              | 40                       | 30    |
| SL5   | 242                 | 3.82              | 23                       | 12    |
| SL1   | 233                 | 7.35              | 17                       | 5     |
| SL3   | 229                 | 11.04             | 23                       | 9     |

The 24 hold-out routes operate between 134 and 220 trips per day, with lengths between 2.66 and 16.44 km. In total, the training routes account for 1,616 daily trips and the hold-out routes for 4,279.

### 3.2 Energy and emission model

Each agent controls a dimensionless speed factor  $s_i$  applied to the scheduled running profile, where  $s_i=1$  denotes the timetable. The energy of one trip is modelled as the sum of traction, congestion-related and auxiliary components:

$$E_i(s) = \kappa L_i (s^2 + c s) + P_{\text{aux}} \frac{T_i}{60 s} \quad (1)$$

where  $\kappa$  is the traction intensity of a battery-electric bus at timetable speed (kWh/km),  $c$  is the fractional congestion overhead, and  $P_{\text{aux}}$  is the auxiliary power (kW) drawn over the running time  $T_i/s$  (min). The quadratic dependence on  $s$  is a stylised representation of the growth of resistive losses with speed, and the auxiliary term captures the opposing effect that slower operation lengthens the time over which heating, ventilation and onboard systems draw power.

For a set of routes  $R$  (the training or the hold-out routes), daily energy and emissions follow from the trip counts and the grid emission factor  $E F_{\text{grid}}$ :

$$E_{\text{tot}} = \sum_{i \in R} n_i E_i(s_i), C O_2 = E F_{\text{grid}} E_{\text{tot}} \quad (2)$$

The grid emission factor is set to 0.2708 kg CO<sub>2</sub>/kWh, which corresponds to the 2024 average emission rate of 597 lb/MWh reported for New England generation (ISO New England, 2025). Because  $E F_{\text{grid}}$  is held constant, the relative reduction in CO<sub>2</sub> is identical to the relative reduction in energy, and the two are reported as a single indicator:

$$\Delta E \% = \Delta C O_2 \% = \left( 1 - \frac{E_{\text{tot}}(s)}{E_{\text{tot}}(1)} \right) \times 100 \% \quad (3)$$

Energy efficiency is normalised by passenger-kilometres, computed as the sum over trips of load times distance rather than as the product of total passengers and total distance:

$$E_{\text{norm}} = \frac{\sum_{i \in R} n_i E_i(s_i)}{\sum_{i \in R} n_i p_{\text{avg}} L_i} \quad (4)$$

GTFS does not contain ridership, so the average load  $p_{\text{avg}}$  is set to 20 passengers per trip for all routes. This value is an assumption rather than an observation;  $E_{\text{norm}}$  scales inversely with  $p_{\text{avg}}$ , whereas  $\Delta E\%$  and OTP do not depend on it.

### 3.3 Service-quality model

The arrival deviation at stop  $j$  of route  $i$  combines the delay accumulated by operating at speed factor  $s_i$  with a stochastic disturbance:

$$\Delta t_{ij} = (t_{ij} - t_{i1}) \left( \frac{1}{s_i} - 1 \right) + \xi_{ij} \quad (5)$$

The first term grows with the scheduled elapsed time since the first stop, so that a given speed reduction produces larger deviations further along the route. The disturbance  $\xi_{ij}$  is the sum of a bias term  $N(0, 1.0^2)$ , a non-negative dwell term  $\max\{0, N(1.5, 0.7^2)\}$  and a congestion term  $N(0, 1.2^2)$ , all in minutes and drawn independently per stop. This surrogate is not calibrated against automatic vehicle location records, and its systematic late bias (a mean dwell excess of 1.5 minutes) is one reason why baseline OTP is near 71%. On-time performance is then

$$OTP = \frac{\sum_{i \in R} n_i \frac{1}{m_i N_{\text{ev}}} \sum_{j=1}^{m_i} \sum_{r=1}^{N_{\text{ev}}} 1(\left| \Delta t_{ij}^{(r)} \right| \leq \delta_{\text{max}})}{\sum_{i \in R} n_i} \times 100\% \quad (6)$$

with  $\delta_{\text{max}} = 2.5$  minutes,  $m_i$  the number of stops, and  $\Delta t_{ij}^{(r)}$  the deviation under the  $r$ -th of  $N_{\text{ev}} = 200$  independent disturbance draws. For training, the indicator is replaced by a smooth surrogate, expressed as a fraction and estimated from  $B$  disturbance draws ( $B = 64$  during training, and 4,000 draws for the baseline value at  $s = 1$ , see Section 3.4),  $\widetilde{OTP}_i(s) = \frac{1}{B m_i} \sum_{b=1}^B \sum_{j=1}^{m_i} \sigma\left(\left(\delta_{\text{max}} - \left| \Delta t_{ij}^{(b)} \right| \right) / \tau\right)$ , where  $\sigma$  is the logistic function and  $\tau = 0.5$  minutes, so that the objective is differentiable in  $s$ .

### 3.4 Policy and local objective

All learning regimes share the same policy class. The speed factor of route  $i$  is a bounded function of a linear score:

$$s_i = 1 + a(z_i) \tanh(z_i), z_i = \theta^\top \phi_i \quad (7)$$

where  $a(z)=0.32$  for  $z<0$  and  $a(z)=0.10$  otherwise, so that  $s_i \in (0.68, 1.10)$ , and  $\phi_i = [1, L_i/20, T_i/60, n_i/100, m_i/50]^T$  is a vector of route descriptors available from GTFS alone. A shared parameter vector  $\theta$  therefore induces route-specific speed factors and can be applied to routes that did not take part in training.

The local objective of agent  $i$  minimises normalised energy subject to a soft constraint on punctuality loss:

$$\ell_i(s) = \frac{E_i(s)}{E_i(1)} + \rho \left[ \max \left( 0, \widetilde{OT\bar{P}}_i(1) - \frac{\varepsilon_{OTP}}{100} - \widetilde{OT\bar{P}}_i(s) \right) \right]^2 + \mu [\max(0, s - 1)]^2 \quad (8)$$

where  $\varepsilon_{OTP}$  is the punctuality tolerance chosen by the operator, in percentage points (2 in the main experiment),  $\rho=200$  is the penalty weight, and  $\mu=0.2$  discourages operation above timetable speed. The baseline  $\widetilde{OT\bar{P}}_i(1)$  is estimated locally by each agent from 4,000 disturbance draws, so that no agent requires information about other routes. Formulating the trade-off as a tolerance, rather than as a weighted sum, makes the operating point interpretable for operators and allows the full trade-off to be traced by varying a single parameter.

### 3.5 Learning regimes

The gradient of  $\ell_i$  with respect to  $s$  is estimated by a central finite difference with common random numbers. At each step,  $B=64$  disturbance draws  $\xi^{(1:B)}$  are sampled once, the loss  $\hat{\ell}_i$  is evaluated with the soft OTP averaged over these draws, and the same draws are used at  $s+h$  and  $s-h$ :

$$\frac{\widehat{\partial \ell_i}}{\partial s} = \frac{\hat{\ell}_i(s+h; \xi^{(1:B)}) - \hat{\ell}_i(s-h; \xi^{(1:B)})}{2h} \quad (9)$$

with  $h=2 \times 10^{-3}$ . The parameter gradient follows by the chain rule,  $\nabla_{\theta} \ell_i = (\partial \ell_i / \partial s) a(z_i) (1 - \tanh^2 z_i) \phi_i$ . All regimes use stochastic gradient descent with the step size  $\eta_k = \eta_0 / (1 + k/\kappa_{\eta})$ ,  $\eta_0=0.004$  and  $\kappa_{\eta}=50$ , start from  $\theta=0$  (that is, from the timetable), and perform  $Q=5$  local steps per agent in each of  $K=400$  rounds, so that the number of gradient evaluations per agent is identical across regimes.

1. **Centralised.** A single  $\theta$  is updated with the trip-weighted average of all agents' gradients, which presupposes that a central entity has access to every route's data. It serves as an upper reference without privacy.
2. **Non-federated MARL.** Each agent updates its own  $\theta_i$  using only its own data and never communicates. No model exists for routes outside the training set, which therefore remain at timetable operation.

3. **FMA-AI (FedAvg).** In each round, every agent starts from the global  $\theta_g^{(k)}$ , performs  $Q$  local steps on its own objective, and returns its parameters, which are aggregated with weights proportional to daily trips:

$$\theta_g^{(k+1)} = \sum_{i=1}^M \frac{n_i}{\sum_{j=1}^M n_j} \theta_i^{(k)} \quad (10)$$

Only the five-dimensional parameter vector is exchanged. Federated averaging limits raw-data exposure but does not by itself provide formal privacy guarantees; secure aggregation or differential privacy would be required for that, and neither is implemented here.

### 3.6 Convergence criterion

Because the gradients are stochastic, the change between consecutive parameters does not vanish monotonically. Convergence is therefore assessed on a moving average over  $W=20$  rounds:

$$R_{\text{conv}} = \min \left\{ k \left| \frac{1}{W} \sum_{q=k-W+1}^k \|\theta^{(q)} - \theta^{(q-1)}\|_2 < \delta_\theta \right. \right\} \quad (11)$$

with  $\delta_\theta = 2 \times 10^{-3}$ , where  $\theta^{(k)}$  denotes the parameters of the regime after round  $k$ : the shared vector for the centralised regime,  $\theta_g^{(k)}$  for FedAvg, and the stacked vectors of all agents for the non-federated regime. In the last case the norm is divided by  $\sqrt{M}$ , with  $M=6$  training agents, so that the criterion is comparable to that of a single shared model. Training always continues to  $K=400$  rounds;  $R_{\text{conv}}$  records when the criterion was first met.

### 3.7 Evaluation protocol

Each experiment is repeated for 20 seeds. Within a seed, all regimes face the same evaluation disturbances (200 independent draws per stop and route), which are generated separately from the training noise. Indicators are computed with the hard OTP of Eq. (6) and reported as means with 95% confidence intervals based on the Student  $t$  distribution; differences between regimes are assessed as paired differences across seeds. A sensitivity analysis repeats the training for  $\varepsilon_{OTP} \in \{0, 1, 2, 3, 5\}$  percentage points with five separate seeds. Table 3 lists all constants.

**Table 3. Constants and parameters used in the implementation**

| Item            | Symbol / setting             | Value                 | Role                             |
|-----------------|------------------------------|-----------------------|----------------------------------|
| GTFS archive    | MBTA, Summer 2025, version D | 31 Jul to 23 Aug 2025 | Routes, trips, shapes, schedules |
| Service day     | Wednesday                    | 6 Aug 2025            | Active services and trip counts  |
| Training agents | $M$                          | 6                     | Most frequent bus routes         |
| Hold-out routes |                              | 24                    | Generalisation test              |

| Item                  | Symbol / setting            | Value                                                                     | Role                            |
|-----------------------|-----------------------------|---------------------------------------------------------------------------|---------------------------------|
| Traction intensity    | $\kappa$                    | 1.60 kWh/km                                                               | Eq. (1)                         |
| Congestion overhead   | $c$                         | 0.15                                                                      | Eq. (1)                         |
| Auxiliary power       | $P_{\text{aux}}$            | 3.0 kW                                                                    | Eq. (1)                         |
| Grid emission factor  | $E F_{\text{grid}}$         | 0.2708 kg CO <sub>2</sub> /kWh                                            | Eq. (2); ISO New England (2025) |
| Average load          | $p_{\text{avg}}$            | 20 passengers per trip                                                    | Eq. (4); assumption             |
| On-time tolerance     | $\delta_{\text{max}}$       | 2.5 min                                                                   | Eq. (6)                         |
| Disturbance model     | bias, dwell, congestion     | $N(0, 1)$ ,<br>$\max\{0, N(1.5, 0.7^2)\}$ ,<br>$N(0, 1.2^2)_{\text{min}}$ | Eq. (5); synthetic              |
| Surrogate temperature | $\tau$                      | 0.5 min                                                                   | Smooth OTP                      |
| Speed-factor range    |                             | 0.68 to 1.10                                                              | Eq. (7)                         |
| Punctuality tolerance | $\epsilon_{\text{OTP}}$     | 2 pp (sweep 0 to 5)                                                       | Eq. (8)                         |
| Penalty weights       | $\rho, \mu$                 | 200, 0.2                                                                  | Eq. (8)                         |
| Gradient estimator    | $h, B$                      | $2 \times 10^{-3}$ , 64                                                   | Eq. (9)                         |
| Optimisation          | $\eta_0, \kappa_\eta, Q, K$ | 0.004, 50, 5, 400                                                         | Section 3.5                     |
| Convergence           | $\delta_\theta, W$          | $2 \times 10^{-3}$ , 20                                                   | Eq. (11)                        |
| Evaluation draws      | $N_{\text{ev}}$             | 200 per stop and route                                                    | Eq. (6)                         |
| Seeds                 |                             | 20 (main), 5 (sweep)                                                      | Section 3.7                     |

The experimental pipeline instantiates part of a broader layered architecture, summarised in Table 4. Layers 1, 2, 3 and 5 are implemented in the present study in the reduced form described above; Layers 4 and 6, as well as the governance functions discussed in Section 5, are architectural extensions that were not evaluated.

**Table 4. Federated Multi-Agent AI architecture for sustainable public transport (FMA-AI-PT)**

| Layer   | Main functionality                 | AI module(s)                                               | Scope in this study                                                                                    |
|---------|------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Layer 1 | Data acquisition and preprocessing | None                                                       | Implemented: GTFS ingestion, service-day resolution, shape-based lengths                               |
| Layer 2 | Local multi-agent control          | Route-level learning agents                                | Implemented: speed-factor policy per route                                                             |
| Layer 3 | Federated policy aggregation       | Federated learning (synchronous FedAvg)                    | Implemented: trip-weighted parameter averaging                                                         |
| Layer 4 | Energy-aware vehicle control       | Eco-driving (acceleration smoothing, regenerative braking) | Not evaluated                                                                                          |
| Layer 5 | Performance evaluation             | Statistical analysis                                       | Implemented: $\Delta E$ , CO <sub>2</sub> , $E_{\text{norm}}$ , OTP, convergence, confidence intervals |
| Layer 6 | Continuous adaptation              | Federated retraining under concept drift                   | Not evaluated                                                                                          |

The model was implemented as a single MATLAB script that uses only core functions, and all indicators reported below were generated in MATLAB. Paired differences between regimes, convergence ranges and the published versions of Figures 1 and 2 were computed in Python from the per-seed outputs of that script.

The same script also runs unchanged in GNU Octave 8.4; an independent replication there, which draws different random-number streams, produced the same route set and indicators whose confidence intervals overlap with those reported here.

## 4. Results

### 4.1 Main comparison

Table 5 reports the indicators for the training routes and the hold-out routes at a punctuality tolerance of 2 percentage points. Timetable operation yields an OTP of 71.2% on the training routes and 71.1% on the hold-out routes, with a daily energy use of 25,682 kWh (6.95 t CO<sub>2</sub>) and 70,194 kWh (19.01 t CO<sub>2</sub>) respectively.

**Table 5. Performance of the three learning regimes ( $\epsilon_{OTP} = 2$  pp; mean  $\pm$  95% CI over 20 seeds)**

| Regime                | Routes   | $\Delta E = \Delta CO_2$<br>(%) | OTP (%)        | E_norm<br>(kWh/pkm) | CO <sub>2</sub> (t/day) | R_conv<br>(median) | Raw data<br>shared |
|-----------------------|----------|---------------------------------|----------------|---------------------|-------------------------|--------------------|--------------------|
| Timetable<br>baseline | Training | 0                               | 71.2           | 0.1000              | 6.95                    |                    |                    |
| Centralised           | Training | $0.83 \pm 0.08$                 | $70.0 \pm 0.3$ | 0.0992              | 6.90                    | 143                | Yes                |
| Non-federated<br>MARL | Training | $0.90 \pm 0.12$                 | $70.0 \pm 0.3$ | 0.0991              | 6.89                    | Not met by<br>400  | No                 |
| FMA-AI<br>(FedAvg)    | Training | $0.78 \pm 0.07$                 | $70.1 \pm 0.3$ | 0.0992              | 6.90                    | 215                | No                 |
| Timetable<br>baseline | Hold-out | 0                               | 71.1           | 0.1024              | 19.01                   |                    |                    |
| Centralised           | Hold-out | $0.80 \pm 0.06$                 | $69.5 \pm 0.1$ | 0.1016              | 18.86                   |                    | Yes                |
| Non-federated<br>MARL | Hold-out | 0 (no model)                    | 71.1           | 0.1024              | 19.01                   |                    | No                 |
| FMA-AI<br>(FedAvg)    | Hold-out | $0.88 \pm 0.06$                 | $69.3 \pm 0.1$ | 0.1015              | 18.84                   |                    | No                 |

On the training routes, the three regimes reach similar operating points. The paired difference between FedAvg and the centralised optimiser is  $-0.05 \pm 0.11$  percentage points in energy and  $+0.07 \pm 0.17$  points in OTP, and neither interval excludes zero. The route-isolated learners save slightly more energy on their own routes than FedAvg (paired difference  $0.12 \pm 0.11$  points, an interval that narrowly excludes zero), while their OTP is  $0.07 \pm 0.15$  points lower. This ordering was not reproduced in the Octave replication mentioned in Section 3.7, and the margin is small relative to the savings themselves, so it should not be over-interpreted. Across the three regimes, the learned speed factors average between 0.994 and 0.995, a reduction of about 0.5% to 0.6% relative to the timetable.

The regimes differ on the hold-out routes. The route-isolated learners have no transferable model and leave these routes at timetable operation, whereas the shared policies learned centrally and by FedAvg reduce energy by 0.80% and 0.88% respectively. Relative to the centralised policy, FedAvg saves  $0.08 \pm 0.08$  points more energy at a cost of  $0.18 \pm 0.17$  points of OTP. Both intervals narrowly exclude zero, but the two differences point in opposite directions and are of the order implied by the trade-off in Section 4.2, which suggests a slightly different position on the same frontier rather than a qualitative difference.

## 4.2 Energy and punctuality trade-off

Table 6 and Figure 1 report the tolerance sweep. With  $\varepsilon_{OTP}=0$ , all regimes slightly increase speed, improving OTP to about 71.5% at the cost of 0.4% to 0.5% more energy. Energy savings appear only when some loss of punctuality is admitted, and they grow approximately linearly with the tolerance: between the tolerances of 1 and 5 percentage points, the ratio of the change in energy reduction to the change in OTP is about 0.66 for both the centralised and the federated frontiers. The federated frontier lies within the confidence intervals of the centralised one at every tolerance. Because each point of the sweep rests on five seeds, its confidence intervals are, for tolerances of 1 percentage point and above, wider than those of the main experiment.

**Table 6. Tolerance sweep (mean  $\pm$  95% CI over 5 seeds; training routes)**

| $\varepsilon_{OTP}$ (pp) | Centralised $\Delta E$ (%) | Centralised OTP (%) | Non-federated $\Delta E$ (%) | Non-federated OTP (%) | FedAvg $\Delta E$ (%) | FedAvg OTP (%) |
|--------------------------|----------------------------|---------------------|------------------------------|-----------------------|-----------------------|----------------|
| 0                        | $-0.43 \pm 0.08$           | $71.5 \pm 0.3$      | $-0.52 \pm 0.05$             | $71.6 \pm 0.3$        | $-0.43 \pm 0.07$      | $71.5 \pm 0.3$ |
| 1                        | $0.29 \pm 0.20$            | $70.5 \pm 0.5$      | $0.17 \pm 0.14$              | $70.7 \pm 0.4$        | $0.17 \pm 0.21$       | $70.7 \pm 0.6$ |
| 2                        | $1.00 \pm 0.19$            | $69.5 \pm 0.4$      | $0.94 \pm 0.18$              | $69.8 \pm 0.6$        | $0.89 \pm 0.30$       | $69.7 \pm 0.7$ |
| 3                        | $1.66 \pm 0.18$            | $68.5 \pm 0.6$      | $1.75 \pm 0.30$              | $68.6 \pm 0.9$        | $1.60 \pm 0.29$       | $68.6 \pm 0.7$ |
| 5                        | $2.88 \pm 0.19$            | $66.6 \pm 0.6$      | $3.21 \pm 0.44$              | $66.3 \pm 1.2$        | $2.89 \pm 0.29$       | $66.6 \pm 0.9$ |

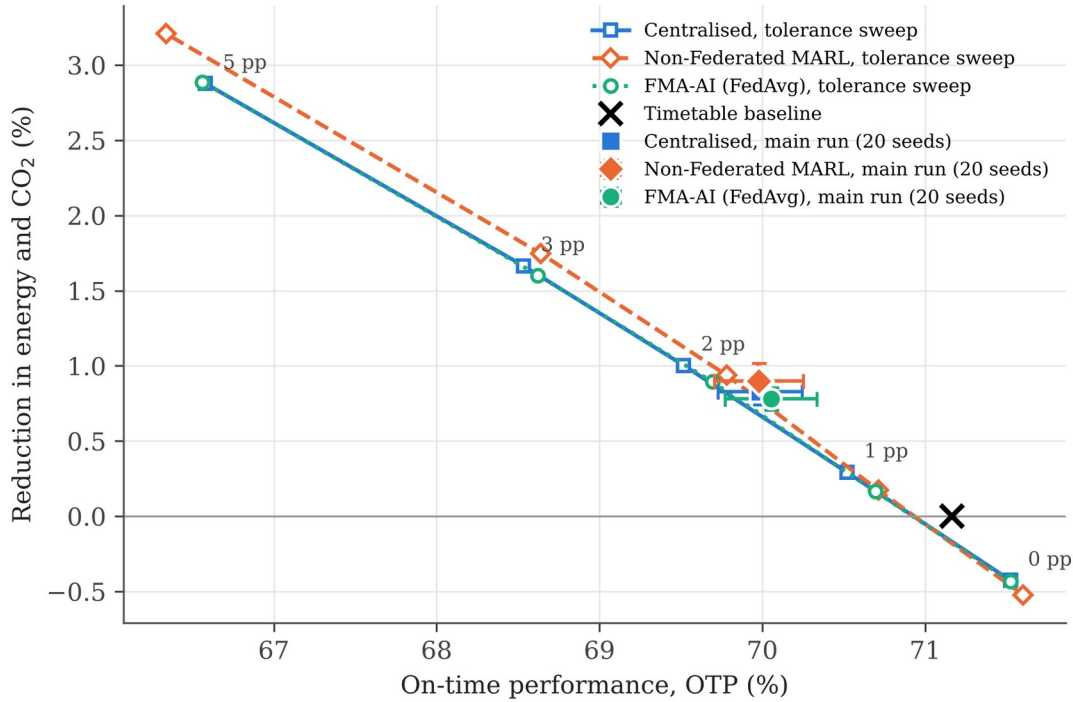


Figure 1. Energy and punctuality trade-off on the training routes. Lines with open markers trace the tolerance sweep (5 seeds per point, labelled by  $\varepsilon_{OTP}$ ); filled markers with 95% confidence intervals show the main experiment at  $\varepsilon_{OTP} = 2$  pp (20 seeds). The two sets use different seeds, which accounts for the small offset between them at 2 pp. The cross marks timetable operation.

### 4.3 Convergence

Figure 2 shows the mean parameter change per round. The centralised optimiser met the criterion of Eq. (11) after a median of 143 rounds (range 89 to 185), and FedAvg after a median of 215 rounds (range 159 to 246). The additional rounds required by FedAvg are consistent with client drift: during the five local steps, each agent moves towards its own optimum before aggregation. The route-isolated learners did not meet the criterion within 400 rounds in any seed. Since each of them estimates five parameters from a single route, its parameters are only weakly identified and continue to wander along directions that barely affect the objective, even though the resulting speed factors, and hence the indicators, are close to those of the other regimes.

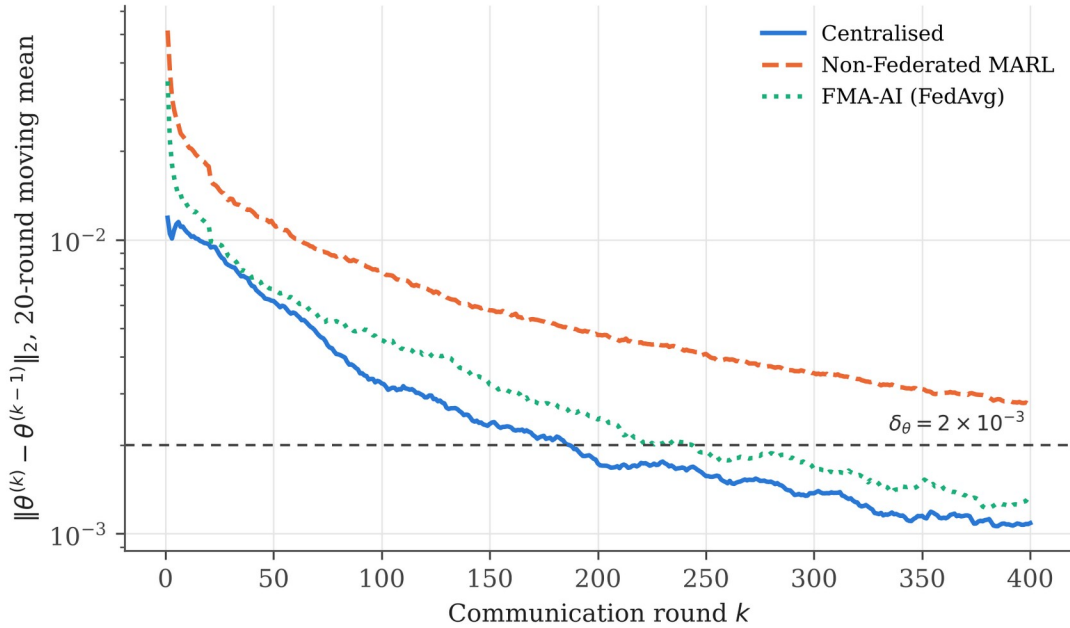


Figure 2. Mean change in model parameters per communication round (20-round moving mean, averaged over 20 seeds). The dashed line marks the convergence threshold  $\delta_\theta$ .

## 5. Discussion

The results support a narrower but more defensible claim than is common in this literature. On the training routes, federated averaging reproduces, within statistical uncertainty, the outcome of a centralised optimiser that pools the data of all routes, while exchanging only a five-dimensional parameter vector per round. Its distinctive advantage over route-isolated learning lies not on the training routes, where isolated learners fitted to a single route perform comparably or marginally better, but in generalisation: the shared policy can be applied to routes that contributed no data, which an operator could exploit to extend a policy to new depots or services without first collecting and centralising their records.

The magnitude of the energy savings is modest. Under the adopted models, speed regulation alone appears to offer roughly 0.66 percentage points of energy per percentage point of punctuality conceded, and no saving

at all without some loss of punctuality. This is a plausible consequence of the model structure: the quadratic traction term rewards slower operation, whereas the delay term in Eq. (5) penalises it in direct proportion to the elapsed schedule, and the late-biased disturbance means that the timetable is already slightly too tight for the simulated conditions. Larger savings would plausibly require control levers that do not lengthen running times, such as smoother acceleration profiles, optimised regenerative braking or holding strategies at timing points, which correspond to Layer 4 of the architecture and are left for future work.

Several limitations qualify these findings. First, the arrival disturbances are synthetic, independent across stops and uncalibrated; real delays are autocorrelated along a route, and calibration against automatic vehicle location data would be required before the absolute OTP levels can be interpreted. Second, the average load of 20 passengers per trip is an assumption, and route-level ridership data would change  $E_{\text{norm}}$ , although not  $\Delta E$  or OTP. Third, the energy model is stylised and treats all vehicles as battery-electric, whereas the MBTA bus fleet is mixed. Fourth, the analysis covers one service day of one feed and six training agents; scaling to tens or hundreds of agents and to heterogeneous fleets may increase client drift, in which case drift-corrected aggregation (Karimireddy et al., 2020) would be a natural extension. Fifth, the privacy benefit is limited to the absence of raw-data transfer; formal guarantees would require secure aggregation or differential privacy.

Table 7 relates the gaps that motivate this study, as set out in Sections 1 and 2 together with two that the analysis brought into focus (transfer of policies across routes and operator governance), to the features of the framework and to the outcomes observed.

**Table 7. Gap-to-feature resolution matrix**

| Gap in literature                                                         | FMA-AI feature                                                         | Operational mechanism                                     | Outcome in the MBTA-based simulation                                                                                                                     |
|---------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Privacy-preserving coordination in transit optimisation is underdeveloped | Synchronous FedAvg with trip-weighted aggregation; no raw data sharing | Agents train locally and transmit only model parameters   | Performance indistinguishable from centralised optimisation (paired $\Delta E$ difference $-0.05 \pm 0.11$ pp); convergence after a median of 215 rounds |
| Limited transfer of learned policies across routes                        | Shared policy over GTFS-derived route descriptors                      | One parameter vector induces route-specific speed factors | 0.88% energy reduction on 24 unseen routes, where isolated learners cannot act                                                                           |
| Limited use of real network data in MARL and FL studies                   | GTFS pipeline with service-day resolution and shape-based lengths      | Schedules, trip counts and geometries define the agents   | Network constructed from the MBTA feed; disturbances remain synthetic                                                                                    |
| Energy and emissions treated as post hoc metrics                          | Energy-minimising objective with an explicit punctuality tolerance     | Penalty on OTP loss beyond the operator’s tolerance       | 0.78% reduction at 2 pp; about 0.66 pp energy per pp OTP across the sweep                                                                                |
| Limited human-in-the-loop governance                                      | Operator-set tolerance $\epsilon_{OTP}$ as the governance lever        | Operators choose the operating point on the frontier      | Proposed; the frontier provides the information an operator would need                                                                                   |

Although the case study is set in a North American network, the architecture is intended to be adaptable. Table 8 maps selected layers to the parameters that are likely to require recalibration in other regions. These mappings are conceptual and were not tested.

**Table 8. Cross-regional adaptation framework for FMA-AI**

| <b>FMA-AI layer</b>                         | <b>Regional adaptation parameter</b>            | <b>European context</b>                                           | <b>Asian context</b>                                                       | <b>Global South context</b>                                                                 |
|---------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Layer 1: Data acquisition                   | Feed format, completeness, temporal granularity | Mature NeTEx and SIRI standards with frequent updates             | Mix of GTFS and proprietary formats; uneven coverage outside megacities    | Sparse GTFS adoption; reliance on mobile GPS, crowdsourcing and community mapping           |
| Layer 4: Energy-aware vehicle control       | Fleet composition, grid emission factor         | Electrified or hybrid fleets; low and declining emission factors  | Mixed fleets; efficiency constrained by congestion and legacy vehicles     | Diesel-dominated fleets and high emission factors; emphasis on basic fuel efficiency        |
| Cross-cutting: Human-in-the-loop governance | Stakeholder structure and governance priorities | Strong unions; EU frameworks embedding climate and social targets | Public-private partnerships; central oversight of AI adoption and data use | Community-led governance and donor-funded projects; emphasis on affordability and inclusion |

In European settings, the emphasis would plausibly fall on high-fidelity NeTEx and SIRI feeds and on low-carbon grids, which reduce the emission benefit of any given energy saving. In Asian megacities, heterogeneous fleets and mixed data formats would mainly affect Layer 1 and the energy model. In the Global South, sparse GTFS coverage and diesel fleets would require alternative data sources and a fuel-based energy model, whereas the federated layer itself would remain unchanged.

## 6. Conclusion

This study evaluated a federated multi-agent framework for energy-aware speed regulation in public transport, using a simulation built on the MBTA GTFS feed. Three findings emerge. First, synchronous federated averaging matched the performance of a centralised optimiser within statistical uncertainty, reducing energy and CO<sub>2</sub> by 0.78% at a punctuality loss of about 1.1 percentage points, without any exchange of operational data. Second, the shared policy generalised to 24 routes that did not participate in training, an outcome that route-isolated learning cannot provide. Third, the attainable savings from speed regulation alone were modest and traded off approximately linearly against punctuality, with no saving available at zero punctuality loss under the adopted assumptions.

These results suggest that the principal value of federation in transit operations may lie in enabling shared, transferable policies across organisational boundaries, rather than in improving performance on any single route. Calibrating the disturbance model against vehicle location data, incorporating ridership, extending the action space to eco-driving and holding control, and adding formal privacy mechanisms are the natural next steps towards a field-ready system.

## Data and code availability

The analysis uses the public MBTA GTFS archive (Summer 2025, version D). The MATLAB script that generates all indicators, the per-seed results, the GNU Octave replication, and the Python script used for the paired statistics and the published figures are openly available on Zenodo (Sargiotis, 2026; <https://doi.org/10.5281/zenodo.23242130>) and GitHub (<https://github.com/uomoathens/fma-ai-mbta>).

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